📐 ENGINEERING MATHEMATICS · B.TECH 1ST YEAR

Linear Differential Equations

Higher Order · Constant Coefficients

UNIT II — Complete Formula & Revision Sheet

CF · PI · Wronskian · Variation of Parameters
Simultaneous DEs · LCR Circuits · SHM

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CF Methods
3 Root Cases

⚡
PI Formulas
5 Standard Forms

🔄
Var. of Params
Step-by-Step

🧩
Wronskian
Linear Indep.

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Simult. DEs
Elimination

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LCR + SHM
Applications

Last-Day Revision Weapon | 25+ Flashcards | Exam Tips | Quick Reference

🧮 KEY DEFINITIONS

Term	Definition	Example / Note
Order	Highest derivative present in the DE	d^3y/dx^3 → Order 3
Degree	Power of highest-order derivative (after clearing radicals/fractions)	$(y'')^2$ → Degree 2
Linear DE	Dependent variable and derivatives appear to the first power only; no products of them	$y'' + 2y' + y = x$
Homogeneous	RHS = 0	$f(D)y = 0$
Non-Homogeneous	RHS ≠ 0 (RHS is function of x)	$f(D)y = X(x)$
CF (Complementary Function)	General solution of $f(D)y = 0$ (homogeneous part)	Contains n arbitrary constants
PI (Particular Integral)	Any particular solution of $f(D)y = X(x)$	No arbitrary constants
General Solution	**y = CF + PI**	**Always: GS = CF + PI**

📐 GENERAL FORM OF LINEAR DE (CONSTANT COEFFICIENTS)

$a_n \cdot (d^ny/dx^n) + a_{n-1} \cdot (d^{n-1}y/dx^{n-1}) + \cdots + a_1 \cdot (dy/dx) + a_0 \cdot y = X(x)$ Using D-operator notation: $(a_nD^n + a_{n-1}D^{n-1} + \cdots + a_1D + a_0)\ y = X(x)$ → Written as: $f(D)\ y = X(x)$ Where $D = d/dx$, $D^2 = d^2/dx^2$, etc.

⚙ AUXILIARY EQUATION (A.E.)

How to Form A.E.:
Replace **D → m** in f(D) = 0

Example:
$f(D) = D^2 + 3D + 2$
A.E.: $m^2 + 3m + 2 = 0$
→ $(m+1)(m+2) = 0$
→ $m = -1, -2$

Types of Roots:

- **Distinct Real:** $m_1 \neq m_2$ (both real)
- **Repeated Real:** $m_1 = m_2$ (equal roots)
- **Complex:** $m = \alpha \pm i\beta$
- **Pure Imaginary:** $m = \pm i\beta$ $(\alpha=0)$

📊 COMPLEMENTARY FUNCTION (CF) — ALL CASES

Case	Nature of Roots (A.E.)	CF Formula
Case 1 Distinct Real	$m_1, m_2, m_3, \ldots$ (all different, real)	$CF = C_1e^{m_1x} + C_2e^{m_2x} + C_3e^{m_3x} + \ldots$
Case 2a Repeated ×2	$m_1 = m_2 = m$ (repeated twice)	$CF = (C_1 + C_2x)e^{mx}$
Case 2b Repeated ×3	$m_1 = m_2 = m_3 = m$ (repeated thrice)	$CF = (C_1 + C_2x + C_3x^2)e^{mx}$
Case 3 Complex	$m = \alpha \pm i\beta$	$CF = e^{\alpha x}[C_1\cos(\beta x) + C_2\sin(\beta x)]$
Case 3a Pure Imaginary	$m = \pm i\beta$ $(\alpha = 0)$	$CF = C_1\cos(\beta x) + C_2\sin(\beta x)$
Case 4 Mixed	Some real + some complex	Combine respective CF formulas

⚠ **Memory Trick:** Distinct → Add exponentials | Repeated → Multiply by $(C_1 + C_2x + \ldots)$ | Complex → $e^{\alpha x}$×trig

⚡ PI FORMULA — GENERAL PRINCIPLE

PI = [1/f(D)] · X(x) f(D) is the operator polynomial; we apply it INVERSELY to X(x).

📌 CASE 1 — X(x) = e^(ax)

PI = [1/f(D)] · e^(ax) = e^(ax) / f(a) [provided f(a) ≠ 0] FAILURE RULE: If f(a) = 0 → multiply by x, differentiate denominator: PI = x · e^(ax) / f'(a) [if f'(a) ≠ 0] PI = x^2 · e^(ax) / f''(a) [if f'(a) = 0 too]

📌 CASE 2 — X(x) = sin(ax) or cos(ax)

Replace D^2 → $-a^2$ in f(D): PI = [1/f(D^2)] · sin(ax) → substitute $D^2 = -a^2$ PI = [1/f(D)] · sin(ax) = sin(ax) / f($-a^2$) [for even powers of D] PI = [1/f(D)] · cos(ax) = cos(ax) / f($-a^2$) FAILURE: If f($-a^2$) = 0 → multiply by x, use: PI = x · [1/f'(D)] · sin(ax) or cos(ax) (where f'(D) means differentiate f(D) w.r.t. D, then substitute $D^2 = -a^2$)

📌 CASE 3 — X(x) = x^n (Polynomial)

PI = [1/f(D)] · x^n Method: Expand $[f(D)]^{-1}$ as a power series in D, then operate on x^n term by term. Note: D^k · x^n = 0 for k > n (series terminates automatically) Example: 1/(1+D) · x^n = (1 − D + D^2 − D^3 + …) · x^n

📌 CASE 4 — X(x) = e^(ax) · V(x) [Exponential Shift]

EXPONENTIAL SHIFT THEOREM: [1/f(D)] · e^(ax) · V(x) = e^(ax) · [1/f(D+a)] · V(x) Steps: 1. Shift out e^(ax) → replace D with (D+a) in f(D) 2. Evaluate [1/f(D+a)] · V(x) using standard methods 3. Multiply result by e^(ax)

📌 CASE 5 — X(x) = x · V(x) [Product Form]

[1/f(D)] · x · V(x) = x · [1/f(D)] · V(x) − [f'(D)/f(D)²] · V(x) Practical: Use exponential shift or differentiate as needed.

🚨 PI FAILURE CASES — MODIFICATION RULES

Condition	Problem	Fix
f(a) = 0	Division by zero for e^(ax)	PI = x·e^(ax)/f'(a)
f($-a^2$) = 0	Zero denominator for sin/cos	Multiply by x; differentiate denominator w.r.t. D^2
Both f(a)=0 and f'(a)=0	Root repeated twice	PI = x^2·e^(ax)/f''(a)

📋 Quick PI Summary Table:

X(x)	PI Formula	Failure Fix
e^(ax)	e^(ax)/f(a)	x·e^(ax)/f'(a)
sin(ax)	sin(ax)/f($-a^2$)	Multiply by x
cos(ax)	cos(ax)/f($-a^2$)	Multiply by x
x^n	Expand $[f(D)]^{-1}$ as series	No failure
e^(ax)·V(x)	e^(ax)·[1/f(D+a)]·V(x)	Exponential shift

WRONSKIAN — DEFINITION & USE

Wronskian of y_1, y_2:

$W(y_1, y_2) = | y_1 \ y_2 | \ | y_1' \ y_2' | \ W = y_1y_2' - y_2y_1'$

Wronskian of y_1, y_2, y_3:

$| y_1 \ y_2 \ y_3 | \ W = | y_1' \ y_2' \ y_3' | \ | y_1'' \ y_2'' \ y_3''|$

Condition for Linear Independence:

$W \neq 0$ → Linearly INDEPENDENT
$W = 0$ → Linearly DEPENDENT

Uses:

- Verify CF solutions are independent
- Required in Variation of Parameters
- Checking solution set validity

Abel's Theorem:
$W(x) = W(x_0)\cdot\exp[-\int p(x)\,dx]$
W is either always zero or never zero.

METHOD OF VARIATION OF PARAMETERS

When to use: When standard PI formulas don't work easily, OR when X(x) is *not* one of the standard types (e.g., sec x, tan x, log x, etc.)

For 2nd-order equation:

$y'' + P(x)y' + Q(x)y = R(x)$ Let CF = $C_1y_1 + C_2y_2$ (y_1, y_2 are known independent solutions) Assume PI = $u_1y_1 + u_2y_2$ (u_1, u_2 are functions of x, not constants) Conditions imposed: $u_1'y_1 + u_2'y_2 = 0$ ... (i) $u_1'y_1' + u_2'y_2' = R(x)$ ... (ii) Solving (i) & (ii) by Cramer's Rule: $u_1' = -y_2\cdot R(x) / W$ $u_2' = y_1\cdot R(x) / W$ $u_1 = -\int[y_2\cdot R(x)/W]\,dx$ $u_2 = \int[y_1\cdot R(x)/W]\,dx$ PI = $u_1y_1 + u_2y_2$

VARIATION OF PARAMETERS — STEP-BY-STEP

1. Find CF → get y_1 and y_2 (two independent solutions of homogeneous equation)
2. Calculate Wronskian: $W = y_1y_2' - y_2y_1'$ (must be ≠ 0)
3. Compute: $u_1' = -y_2R(x)/W$ and $u_2' = y_1R(x)/W$
4. Integrate: $u_1 = \int u_1'\,dx$ and $u_2 = \int u_2'\,dx$
5. PI = $u_1y_1 + u_2y_2$ → General Solution = CF + PI

⚠ **Common Mistakes in Variation of Parameters:**

- Forgetting to write equation in standard form (coefficient of y'' = 1) before reading R(x)
- Computing W incorrectly (sign error)
- Integration errors in u_1, u_2
- Mixing up y_1 and y_2 in the formulas for u_1' and u_2'

COMPARISON: D-OPERATOR PI vs VARIATION OF PARAMETERS

Aspect	D-Operator Method	Variation of Parameters
Works for	Standard X(x): e^ax, sin, cos, poly	Any R(x) including sec, tan, log, etc.
Requires	Only the operator f(D)	CF solutions y_1, y_2 first
Complexity	Simpler for standard forms	More work but universally applicable
Failure case	When f(a)=0 or f(−a²)=0 (need modification)	No failure (always works if W≠0)

🔗 SIMULTANEOUS LINEAR DIFFERENTIAL EQUATIONS

Definition: A system of DEs involving two or more dependent variables (x, y) and one independent variable (t or x).

General Form (2 variables x, y; independent variable t): $f_1(D)x + g_1(D)y = X_1(t)$ $f_2(D)x + g_2(D)y = X_2(t)$ where $D = d/dt$

🔧 ELIMINATION METHOD — STEPS

1. Operate on Equation 1 with $f_2(D)$ and on Equation 2 with $f_1(D)$ (or suitable operators)
2. Subtract (or add) to eliminate one variable (say y) → get DE in x only
3. Solve the resulting DE → find x(t) = CF + PI
4. Substitute x(t) back into either original equation → solve for y(t)
5. Note: Constants in x and y are NOT independent — relate them through original equation

⚡ APPLICATION: L-C-R CIRCUIT

Governing DE (charge form): $L\cdot(d^2q/dt^2) + R\cdot(dq/dt) + (1/C)\cdot q = E(t)$ Governing DE (current form): $L\cdot(d^2i/dt^2) + R\cdot(di/dt) + (1/C)\cdot i = dE/dt$ Where: L = Inductance (H) R = Resistance (Ω) C = Capacitance (F) q = Charge (C), $i = dq/dt$ = Current (A) E(t) = EMF (V)

Characteristic Equation:
$Lm^2 + Rm + 1/C = 0$

Discriminant:
$\Delta = R^2 - 4L/C$

Case	Condition	Nature	Solution Behaviour
Overdamped	$R^2 > 4L/C$ ($\Delta > 0$)	2 distinct real roots	Charge decays without oscillation
Critically Damped	$R^2 = 4L/C$ ($\Delta = 0$)	2 equal real roots	Fastest decay to zero without oscillation
Underdamped	$R^2 < 4L/C$ ($\Delta < 0$)	Complex conjugate roots	Oscillatory decay (damped oscillations)
Undamped (LC)	$R = 0$	Pure imaginary roots	Sustained oscillations at resonant freq.

Resonant Frequency: $\omega_0 = 1/\sqrt{LC}$ **Natural Frequency:** $f_0 = 1/(2\pi\sqrt{LC})$

🎯 APPLICATION: SIMPLE HARMONIC MOTION (SHM)

SHM Equation: $d^2x/dt^2 + \omega^2x = 0$ Solution: $x(t) = A\cdot\cos(\omega t) + B\cdot\sin(\omega t)$ OR $x(t) = R\cdot\cos(\omega t - \varphi)$ where $R = \sqrt{A^2+B^2}$, $\tan(\varphi) = B/A$ Angular Frequency: $\omega = \sqrt{k/m}$ [k=spring const, m=mass] Period: $T = 2\pi/\omega$ Frequency: $f = \omega/(2\pi) = 1/T$ Amplitude: $R = \sqrt{A^2+B^2}$

Damped SHM:
$d^2x/dt^2 + 2k\cdot dx/dt + \omega^2x = 0$

Forced SHM:
$d^2x/dt^2 + \omega^2x = F\cdot\sin(pt)$

Resonance: $p = \omega$ (forcing freq = natural freq)
Amplitude → ∞ (theoretically)

Energy in SHM:
$KE = \frac{1}{2}m\cdot\omega^2(A^2-x^2)$
$PE = \frac{1}{2}m\cdot\omega^2x^2$
Total $E = \frac{1}{2}m\cdot\omega^2A^2$ = const

📊 LCR vs SHM — ANALOGY TABLE

LCR Circuit	Mechanical SHM
Inductance L	Mass m
Resistance R	Damping coefficient c
1/Capacitance (1/C)	Spring constant k
Charge q	Displacement x

Current i = dq/dt	Velocity v = dx/dt
EMF E(t)	Applied force F(t)
`Lq'' + Rq' + q/C = E`	`mx'' + cx' + kx = F`

🃏 FLASHCARDS 1-16 (Cover & Test Yourself!)

Q1: General solution of a linear DE consists of?

A: CF + PI (Complementary Function + Particular Integral)

Q2: What is a homogeneous linear DE?

A: RHS = 0 → $f(D)y = 0$

Q3: How is the Auxiliary Equation (A.E.) formed?

A: Replace D with m in $f(D)$, then set $f(m) = 0$

Q4: CF for distinct real roots m_1, m_2, m_3?

A: $C_1e^{m_1x} + C_2e^{m_2x} + C_3e^{m_3x}$

Q5: CF for repeated root m (twice)?

A: $(C_1 + C_2x)e^{mx}$

Q6: CF for repeated root m (three times)?

A: $(C_1 + C_2x + C_3x^2)e^{mx}$

Q7: CF for complex roots $m = \alpha \pm i\beta$?

A: $e^{\alpha x}[C_1\cos(\beta x) + C_2\sin(\beta x)]$

Q8: CF for pure imaginary roots $m = \pm i\beta$?

A: $C_1\cos(\beta x) + C_2\sin(\beta x)$ (no e^x factor)

Q9: PI formula for $X = e^{ax}$?

A: PI = $e^{ax} / f(a)$ [if $f(a) \neq 0$]

Q10: PI failure rule when $f(a) = 0$ for $X = e^{ax}$?

A: PI = $x \cdot e^{ax} / f'(a)$ [multiply by x, diff. denominator]

Q11: PI formula for $X = \sin(ax)$ or $\cos(ax)$?

A: Replace $D^2 = -a^2$ in $f(D)$ → $\sin(ax)/f(-a^2)$

Q12: Exponential Shift Theorem?

A: $[1/f(D)] \cdot e^{ax} \cdot V = e^{ax} \cdot [1/f(D+a)] \cdot V$

Q13: Wronskian $W(y_1, y_2)$ = ?

A: $W = y_1y_2' - y_2y_1'$

Q14: $W \neq 0$ means?

A: y_1, y_2 are Linearly INDEPENDENT (valid CF)

Q15: In Variation of Parameters, u_1' = ?

A: $u_1' = -y_2R(x) / W$

Q16: In Variation of Parameters, u_2' = ?

A: $u_2' = y_1R(x) / W$

🃏 FLASHCARDS 17–30

Q17: SHM governing equation?

A: $d^2x/dt^2 + \omega^2 x = 0$

Q18: General solution of SHM?

A: $x = A\cdot\cos(\omega t) + B\cdot\sin(\omega t)$

Q19: Angular frequency ω in SHM?

A: $\omega = \sqrt{(k/m)}$ [k=spring const, m=mass]

Q20: Time period of SHM?

A: $T = 2\pi/\omega$

Q21: LCR circuit DE (charge form)?

A: $L\cdot q'' + R\cdot q' + q/C = E(t)$

Q22: Condition for underdamped LCR?

A: $R^2 < 4L/C$ (complex roots → oscillatory decay)

Q23: Condition for critically damped LCR?

A: $R^2 = 4L/C$ (equal roots → fastest decay)

Q24: Condition for overdamped LCR?

A: $R^2 > 4L/C$ (distinct real roots → no oscillation)

Q25: Resonant frequency of LC circuit?

A: $\omega_0 = 1/\sqrt{(LC)}$

Q26: In simultaneous DEs, the elimination method does what?

A: Eliminates one variable to get a single DE in one unknown

Q27: How many arbitrary constants in GS of nth order DE?

A: n constants (exactly)

Q28: D-operator notation: D^2y means?

A: d^2y/dx^2 (second derivative of y)

Q29: PI for polynomial x^n — method?

A: Expand $[f(D)]^{-1}$ as binomial/power series, operate on x^n

Q30: When to use Variation of Parameters over D-operator?

A: When X(x) = sec x, tan x, log x, or non-standard function

💡 **Study Strategy:** Cover the answer (yellow box) with your hand, read the question, then reveal. Practice until you can answer within 3 seconds without hesitation!

🚨 MOST COMMON EXAM ERRORS

✕ MISTAKE 1: Confusing CF and PI
CF = solution of homogeneous equation (RHS=0)
PI = solution when RHS = $X(x) \neq 0$
✓ GS always = CF + PI. Never skip PI!

✕ MISTAKE 2: Wrong Auxiliary Equation
Many students forget to SET $f(m) = 0$ after substituting $D \to m$.
✓ A.E.: Replace D with m, equate to ZERO.

✕ MISTAKE 3: Repeated Root CF Error
Common error: writing $C_1e^{mx} + C_2e^{mx}$ for repeated roots.
✓ Correct: $(C_1 + C_2x)e^{mx}$

✕ MISTAKE 4: Ignoring PI Failure Cases
If $f(a) = 0$ for e^{ax} PI, division gives ∞.
✓ Always check f(a) before computing. If zero → multiply by x.

✕ MISTAKE 5: Sin/Cos PI — Wrong Substitution
$D^2 \to -a^2$ (NOT $D \to -a$). Students often substitute $D \to -a$.
✓ Replace $D^2 = -a^2$. Keep D terms as is initially.

✕ MISTAKE 6: Wronskian Sign Error
$W = y_1y_2' - y_2y_1'$ (subtract, don't add)
✓ The determinant formula: top-left×bottom-right MINUS top-right×bottom-left.

✕ MISTAKE 7: Variation of Parameters — Forgetting Standard Form
Must divide entire equation by leading coefficient before reading $R(x)$.
✓ Coefficient of y'' MUST be 1 before applying formulas.

✕ MISTAKE 8: Mixing SHM and LCR Equations
SHM: $d^2x/dt^2 + \omega^2x = 0$ (no first derivative!)
LCR: $Lq'' + Rq' + q/C = E(t)$ (has first derivative = R term)
✓ LCR has the R·q' damping term; SHM doesn't (undamped).

✕ MISTAKE 9: Sign Errors in D-Operator Series
$1/(1+D) = 1 - D + D^2 - D^3 + \ldots$ (alternating signs!)
$1/(1-D) = 1 + D + D^2 + D^3 + \ldots$ (all positive)
✓ Be careful with +D vs −D in the expansion.

✕ MISTAKE 10: Simultaneous DEs — Extra Constants
Students write extra arbitrary constants for y after finding x.
✓ Constants in y are determined by substituting back. Don't add new ones!

✕ MISTAKE 11: Complex Roots — Wrong α and β
For $m = -3 \pm 2i$: $\alpha = -3$, $\beta = 2$
✓ CF = $e^{-3x}[C_1\cos(2x) + C_2\sin(2x)]$. Carefully identify α and β.

✕ MISTAKE 12: Forgetting Integration in Var. of Params.
u_1' and u_2' are derivatives; you must INTEGRATE to get u_1 and u_2.
✓ $u_1 = \int u_1'dx$, $u_2 = \int u_2'dx$ (don't skip integration!)

✅ GOLDEN RULES FOR EXAM SUCCESS

- Always form A.E. first for any linear DE
- Check discriminant to identify root type
- Check $f(a)=0$ BEFORE computing PI for e^{ax}
- For Var. of Params: normalize equation first (y'' coeff = 1)

- GS = CF + PI — NEVER forget to add PI
- Count constants: nth order → n constants in CF
- Verify $W \neq 0$ before using Var. of Params
- SHM period $T = 2\pi/\omega$ (not $2\pi\cdot\omega$)

⚙ GENERAL FORM & SOLUTION STRUCTURE

DE: $f(D)y = X(x)$ | $f(D) = a_n D^n + a_{n-1} D^{n-1} + \cdots + a_0$

GS: $y = CF + PI$ | A.E.: $f(m) = 0$ [replace $D \to m$]

📊 CF — COMPLEMENTARY FUNCTION

Distinct real m_1, m_2: $CF = C_1 e^{(m_1 x)} + C_2 e^{(m_2 x)}$

Repeated m (×2): $CF = (C_1 + C_2 x) e^{(mx)}$

Repeated m (×3): $CF = (C_1 + C_2 x + C_3 x^2) e^{(mx)}$

Complex $\alpha \pm i\beta$: $CF = e^{(\alpha x)}[C_1 \cos(\beta x) + C_2 \sin(\beta x)]$

Pure imaginary $\pm i\beta$: $CF = C_1 \cos(\beta x) + C_2 \sin(\beta x)$

⚡ PI — PARTICULAR INTEGRAL

$e^{(ax)}$: $PI = e^{(ax)}/f(a)$ [failure: $PI = x \cdot e^{(ax)}/f'(a)$]

$\sin(ax)/\cos(ax)$: $PI = \sin/\cos(ax)/f(-a^2)$ [replace $D^2 = -a^2$]

x^n: $PI = [f(D)]^{-1} \cdot x^n$ [expand as power series in D]

$e^{(ax)}V(x)$: $PI = e^{(ax)} \cdot [1/f(D+a)] \cdot V(x)$ [exp. shift]

🔀 WRONSKIAN

$W(y_1, y_2)$: $y_1 y_2' - y_2 y_1'$ | $W \neq 0 \to$ Linearly Independent

🔄 VARIATION OF PARAMETERS (2nd order)

u_1': $-y_2 R/W$ u_2': $y_1 R/W$ PI: $u_1 y_1 + u_2 y_2$

Normalize first: coefficient of $y'' = 1$

⚡ LCR CIRCUIT

DE: $Lq'' + Rq' + q/C = E(t)$

Overdamped: $R^2 > 4L/C$ | Critically: $R^2 = 4L/C$ | Underdamped: $R^2 < 4L/C$

Resonant ω_0: $1/\sqrt{(LC)}$

🌀 SIMPLE HARMONIC MOTION

DE: $x'' + \omega^2 x = 0$ | Solution: $x = A\cos(\omega t) + B\sin(\omega t)$

ω: $\sqrt{(k/m)}$ | T: $2\pi/\omega$ | f: $\omega/(2\pi)$ | Amplitude: $\sqrt{(A^2+B^2)}$

📋 MASTER TABLE — TYPES OF ROOTS → CF FORM

Roots of A.E.	Type	CF	Example A.E.
$m = 2, -3$	Distinct Real	$C_1 e^{(2x)} + C_2 e^{(-3x)}$	$m^2+m-6=0$
$m = 2, 2$	Repeated (×2)	$(C_1+C_2x)e^{(2x)}$	$(m-2)^2=0$
$m = 2, 2, 2$	Repeated (×3)	$(C_1+C_2x+C_3x^2)e^{(2x)}$	$(m-2)^3=0$
$m = 3 \pm 2i$	Complex	$e^{(3x)}[C_1\cos(2x)+C_2\sin(2x)]$	$m^2-6m+13=0$
$m = \pm 3i$	Pure Imaginary	$C_1\cos(3x) + C_2\sin(3x)$	$m^2+9=0$
$m = 0$	Zero root	C (constant)	$m=0$
$m = 0, 0$	Repeated zero	$C_1 + C_2 x$	$m^2=0$

📝 HOW TO SOLVE A LINEAR DE — COMPLETE ALGORITHM

1. **Identify the type:** Is RHS = 0? (Homogeneous) or RHS = X(x)? (Non-homogeneous)
2. **Write in operator form:** Express as $f(D)y = X(x)$ using $D = d/dx$
3. **Form Auxiliary Equation:** Set $f(m) = 0$ (replace D with m)
4. **Solve A.E.:** Find roots $m_1, m_2, \ldots$ Identify: distinct / repeated / complex
5. **Write CF:** Use appropriate CF formula based on root type
6. **Find PI:** Apply appropriate PI formula for the given X(x). Check for failure case!
7. **Write GS:** GS = CF + PI. Apply initial/boundary conditions if given.

🎯 CHOOSING THE RIGHT PI METHOD

X(x) Type	Method	Key Check
`e^(ax)`	Direct: PI = $e^{(ax)}/f(a)$	Check $f(a) \neq 0$
`sin(ax), cos(ax)`	Replace $D^2 = -a^2$	Check $f(-a^2) \neq 0$
x^n, polynomial	Binomial series expansion of $[f(D)]^{-1}$	Expand up to D^n term
`e^(ax)·sin/cos(ax)`	Exponential shift: replace D with (D+a)	Then apply sin/cos rule
$e^{(ax)} \cdot x^n$	Exponential shift: replace D with (D+a)	Then apply polynomial rule
`sec x, tan x, log x`	Variation of Parameters	Must know CF first ($W \neq 0$)
Any product not above	Variation of Parameters	Most general method

🔗 SIMULTANEOUS DE — WORKED FORMAT

Given: $dx/dt + 2y = t$ and $dy/dt - 2x = 1$

Step 1: Write as $(D)x + 2y = t$ and $-2x + (D)y = 1$
Step 2: Operate on Eq.1 with D: $D^2x + 2Dy = 1$
Step 3: From Eq.2: $Dy = 2x + 1$ → substitute: $D^2x + 2(2x+1) = 1$
Step 4: $D^2x + 4x = -1$ → solve for x (standard PI/CF)
Step 5: Find y from original equation

📊 UNIT II — COMPLETE TOPIC MAP

Topic	Sub-Topics	Key Formulas	Marks Weight
CF Methods	3 root cases	CF table	⭐⭐⭐⭐
PI Methods	e^ax, sin/cos, poly, product	5 standard formulas	⭐⭐⭐⭐⭐
Wronskian	2×2 & 3×3 determinant	$W = y_1y_2' - y_2y_1'$	⭐⭐
Variation of Params.	u_1, u_2 formulas	$u_1' = -y_2R/W$, $u_2' = y_1R/W$	⭐⭐⭐⭐
Simultaneous DEs	Elimination method	Operator elimination	⭐⭐⭐
LCR Circuits	3 damping cases	$Lq''+Rq'+q/C=E$	⭐⭐⭐
SHM	Undamped, damped, forced	$x''+\omega^2x=0$	⭐⭐⭐

🎯 FINAL MANTRA FOR EXAM

`Form A.E. → Find Roots → Write CF → Find PI → GS = CF + PI`

Check failure case for PI · Normalize before Variation of Parameters · Constants in Simultaneous DEs are linked